

5.1 Joint Probability Distributions (the discrete case)

When in an experiment several aspects are investigated at the same time, this is referred to as a multidimensional random variable.

Example 1

Selection data for a person's suitability in language (= α) and math (= β) skills:

$$\begin{aligned} X &= \text{score on a } \alpha\text{-test} & (\text{for } x \in \{10, 20, 30\}) \\ Y &= \text{score on a } \beta\text{-test} & (\text{for } y \in \{10, 20, 30\}) \end{aligned}$$

So you can ask about:

$$\begin{aligned} P(X = 20 \text{ and } Y = 30) &= P(X = 20, Y = 30) \\ &= P((X, Y) = (20, 30)) \end{aligned}$$

(X, Y) is called a 2-dimensional discrete random variable.

De function

$$f_{XY}(x, y) = P(X = x, Y = y)$$

is called the joint probability mass function of the discrete random variables X and Y .

Example 2

(see the data in Example 1)

Suppose

$y \backslash x$	10	20	30	
10	0.05	0.05	0.20	0.30
20	0.10	0.20	0.10	0.40
30	0.10	0.15	0.05	0.30
	0.25	0.40	0.35	1

Then

- $P(X \geq 20, Y \geq 20) = 0.20 + 0.10 + 0.15 + 0.05 = 0.50$
- $P(X = Y) = 0.05 + 0.20 + 0.05 = 0.30$
- $P(X = 10) = P((X, Y) = (10, 10) \vee (10, 20) \vee (10, 30))$
 $= \sum_y P(X = 10, Y = y)$
 $= 0.05 + 0.05 + 0.20 = 0.30$

Summing each row creates the marginal probability distribution of X . In a similar way (i.e. summing each column), the marginal probability distribution of Y arises.

Then the expected value and the variance of X and Y can be determined:

- $E(X) = 10 \cdot 0.3 + 20 \cdot 0.4 + 30 \cdot 0.3 = 20$; $E(Y) = 10 \cdot 0.25 + 20 \cdot 0.4 + 30 \cdot 0.35 = 21$
- $V(X) = E(X^2) - E(X)^2 = 10^2 \cdot 0.3 + 20^2 \cdot 0.4 + 30^2 \cdot 0.3 - 20^2 = 460 - 20^2 = 60$; $V(Y) = 59$

Conditional Probability Distributions

$$f_{Y|X}(y) = P(Y = y | X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f_{XY}(x, y)}{f_X(x)}$$

Example 3

(see the data in Example 2)

$$f_{Y|10}(y) = \begin{cases} \frac{0.05}{0.30} = \frac{1}{6} & , \text{ for } y = 10 \\ \frac{1}{6} & , \text{ for } y = 20 \\ \frac{2}{3} & , \text{ for } y = 30 \end{cases}$$

So

$$E(Y|10) = 10 \cdot \frac{1}{6} + 20 \cdot \frac{1}{6} + 30 \cdot \frac{2}{3} = 25$$

Therefore

$$E(Y|10) \neq E(Y) = 21$$

i.e.

X influences Y .

Independence

The discrete random variables X and Y are called independent, if the following applies to all values of x and y :

$$\begin{aligned} P(X = x, Y = y) &= P(X = x) \cdot P(Y = y) \\ \text{or : } P(X = x | Y = y) &= P(X = x) \\ \text{or : } P(Y = y | X = x) &= P(Y = y) \end{aligned}$$

Example 4

(see the data in Example 2)

$$\begin{aligned} P(X = 10, Y = 10) &= 0.05 \\ P(X = 10) \cdot P(Y = 10) &= 0.30 \cdot 0.25 = 0.075 \end{aligned}$$

So X and Y are **not** independent.

5.2 Covariance and Correlation

Given are twelve number pairs (x,y) with a clearly observable **negative relationship** between X and Y :

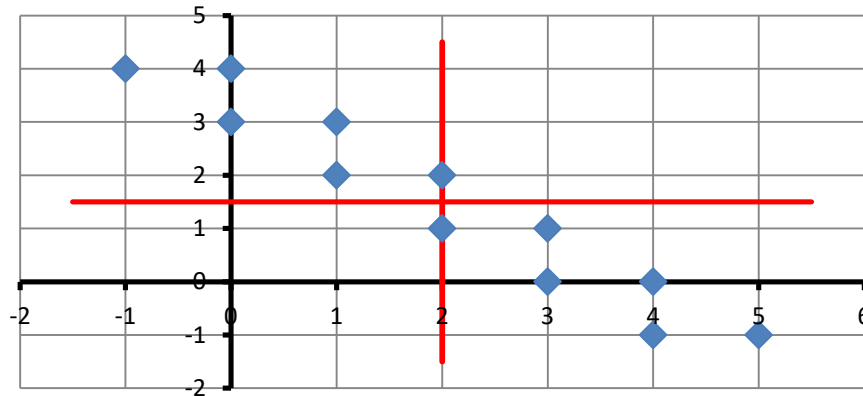


Figure 5.1 Negative dependency

To quantify this relationship, the coordinate system is shifted from $(0, 0)$ to $(\bar{x}, \bar{y}) = (2, 1\frac{1}{2})$. Relative to this new coordinate system, all points are in the quadrants II and IV, i.e. $(-, +)$ and $(+, -)$.

Therefore the **covariance** between the random variables X and Y , denoted as $\text{cov}(X,Y)$ or σ_{XY} , is defined as follows:

$$\text{cov}(X,Y) = E[(X - \mu_X) \cdot (Y - \mu_Y)] = \sigma_{XY}$$

It is not difficult to reduce this expression to:

$$\text{cov}(X,Y) = E(XY) - E(X) \cdot E(Y)$$

Example 5

(see the data in Example 2)

$$\begin{aligned} E(XY) &= (10 \cdot 10) \cdot 0.05 + (10 \cdot 20) \cdot 0.05 + 300 \cdot 0.20 + \dots + 900 \cdot 0.05 \\ &= 400 \end{aligned}$$

$$\begin{aligned} \text{cov}(X,Y) &= E(X \cdot Y) - E(X) \cdot E(Y) \\ &= 400 - 20 \cdot 21 = -20 \end{aligned}$$

This result indicates a **negative relationship** between language and math skills. Because a covariance has no lower or upper limit, it is difficult to make a statement about the strength of this relationship.

There is another measure of the relationship between two random variables that is often easier to interpret than the covariance, namely the **correlation coefficient**. Scaling the covariance by the product of the standard deviation of each variable, a measure between -1 and $+1$ arises.

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}$$

The following result can be shown:

$$-1 \leq \rho_{XY} \leq 1$$

It applies:

- $\rho_{XY} = 0$ $\Leftrightarrow$ no linear dependency
- $\rho_{XY} = \pm 1$ $\Leftrightarrow$ perfect positive/negative linear dependency
(i.e. the points fall exactly along a straight line)

Example 6

(see the data in the Examples 2 and 5)

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}} = \frac{-20}{\sqrt{60 \cdot 59}} = -0.3361$$

So there is a **weak** negative linear dependency between language and math skills.

Theorem

If X and Y are **independent** random variables, then $E(X \cdot Y) = E(X) \cdot E(Y)$.

Proof

$$\begin{aligned} E(X \cdot Y) &= \sum_x \sum_y x \cdot y \cdot P(X = x, Y = y) \\ &= \sum_x \sum_y x \cdot y \cdot P(X = x) \cdot P(Y = y) \\ &= \left(\sum_x x \cdot P(X = x) \right) \cdot \left(\sum_y y \cdot P(Y = y) \right) \\ &= E(X) \cdot E(Y) \end{aligned}$$

Corollary

If X and Y are **independent** random variables, then $\text{cov}(X, Y) = \rho_{XY} = 0$.

Remark

If $\rho_{XY} = 0$, X and Y are called **uncorrelated**. In that case there is no linear relationship between X and Y . The reverse statement (**uncorrelated** $\rightarrow$ **independent**) is **not** true. See Figure 5-12 (d) on page 170.